

# Research and Development of an Automated Concept Drift Detection and Adaptive Machine Learning System

Master's Thesis

Student: Le Phuc Duc

Supervisor: Assoc.Prof.Dr. Thoai Nam

Ho Chi Minh City University of Technology  
Faculty of Computer Science and Engineering

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# Presentation Outline

- 1 Introduction
- 2 Theoretical Background
- 3 Proposed Architecture
- 4 Experiments & Evaluation
- 5 Conclusion

# Motivation & Background

## Real-world examples:

- Banking fraud detectors: false-alarm rate **triples** within a year of deployment.
- Predictive maintenance: model accuracy degrades as machine conditions change.

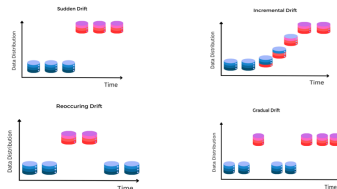
## Underlying problem — Concept Drift:

$$P_{\text{train}}(X, Y) \neq P_{\text{online},t}(X, Y)$$

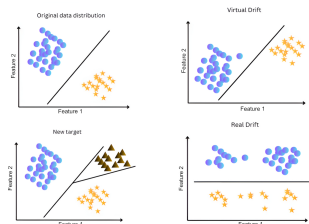
### Non-stationary environments

The mapping between inputs and outputs changes over time.

*Example:* legitimate transactions get flagged as fraud during a seasonal spike because the model's notion of “normal” is now stale.



Hình: Temporal drift patterns.



Hình: Distributional shift in the joint distribution.

# Objectives & Contributions

## Research Objectives

- ➊ **Faster, more reliable detection:** integrate IDW-MMD into ShapeDD to raise throughput while keeping false alarms near zero.
- ➋ **Unsupervised type identification:** build SE-CDT to categorize drift behaviors *without ground-truth labels*.
- ➌ **End-to-end system:** deploy a closed-loop pipeline on Apache Kafka.

## Three Main Contributions

- ➊ **Detection (ShapeDD-IDW):** inverse density weighting + Gamma-null p-value  $\rightarrow$  a calibrated test ( $\alpha = 0.05$ ) with a modest end-to-end speedup over ShapeDD.
- ➋ **Classification (SE-CDT):** an unsupervised geometric analyzer that labels 5 drift types directly from the MMD signal  $\sigma(t)$ .
- ➌ **Adaptation:** a type-aware router that picks an update strategy (full reset, warm-start, model cache, ignore) based on the detected drift type.

# Maximum Mean Discrepancy (MMD)

**Idea:** compare two distributions  $P, Q$  via their mean embeddings in an RKHS.

$$\text{MMD}(P, Q) = \sup_{f \in \mathcal{F}} |\mathbb{E}_{X \sim P}[f(X)] - \mathbb{E}_{Y \sim Q}[f(Y)]|$$

**Empirical estimator** (Gaussian RBF kernel):

$$\widehat{\text{MMD}}^2 = \underbrace{\frac{1}{n^2} \sum_{i,j} k(x_i, x_j)}_{\text{within } P} + \underbrace{\frac{1}{m^2} \sum_{p,q} k(y_p, y_q)}_{\text{within } Q} - \underbrace{\frac{2}{nm} \sum_{i,p} k(x_i, y_p)}_{\text{cross } P-Q}$$

$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma_k^2}\right) \quad (\text{Gaussian RBF})$$

## Role in drift detection

- $\widehat{\text{MMD}}^2 \approx 0$ : distributions look the same.
- $\widehat{\text{MMD}}^2 \gg 0$ : distributions disagree  $\rightarrow$  **drift candidate**.

## Why it suits this problem

Each sample is mapped into an RKHS via the kernel; the test compares the mean embeddings of the two windows. Result: a single scalar test statistic, no distributional assumption, and no labels required at runtime.

# ShapeDD — Shape-Based Drift Detection

## Triangle Shape Theorem:

For a sudden drift at  $t = 0$ , the MMD trace satisfies

$$\sigma(t) = \underbrace{\|P - Q\|}_{\text{drift severity}} \cdot \underbrace{h_l(t)}_{\text{triangular envelope}},$$

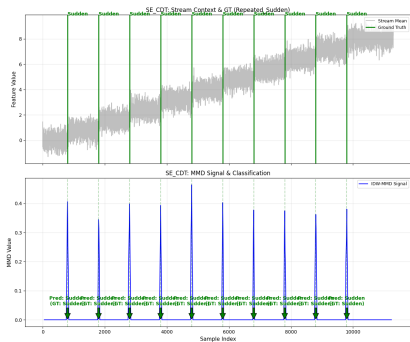
with  $h_l(t) = \max\left(0, 1 - \frac{|l-t|}{l}\right)$ .

## Processing pipeline:

- 1 Maintain a sliding window of  $2l_1$  samples.
- 2 Build the kernel matrix  $K$ .
- 3 Compute MMD between the two halves.
- 4 Test significance with a permutation test.

## Bottleneck

The permutation test re-builds the kernel thousands of times → heavy compute and the null distribution must still be estimated correctly.



**Hình:** Sudden drift produces the predicted triangular MMD trace.

## Why a triangle?

As the sliding window crosses the change point, the proportion of post-drift samples grows linearly → MMD rises linearly until the window is fully on the new distribution, then falls back symmetrically.

# CDT-MSW — Supervised Classification Baseline

CDT-MSW (Guo et al., 2022) groups drift into:

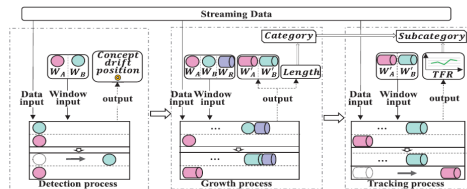
- **TCD (Transient)**: Sudden, Blip, Recurrent.
- **PCD (Progressive)**: Gradual, Incremental.

## Growth Process:

- Tracks the variance of accuracy degradation over a multi-sliding window.
- Short degradation  $\rightarrow$  TCD; sustained degradation  $\rightarrow$  PCD.

## Limitation

CDT-MSW needs **ground-truth labels** at every step to compute the accuracy curve — impractical for IIoT streams where labels arrive late or never.

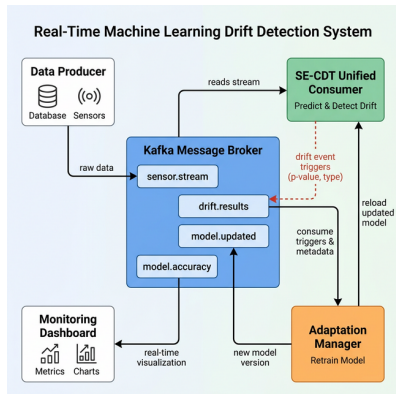


Hình: CDT-MSW architecture (label-dependent).

## Motivation for SE-CDT

Replace the accuracy curve with the *shape* of the MMD signal  $\sigma(t)$ . The same TCD/PCD distinction can then be made **without any labels at runtime**.

# SE-CDT System Architecture



## Layer 1 — Detection

ShapeDD + IDW-MMD

Decides *when* a drift in  $P(X)$  occurs.

## Layer 2 — Classification

SE-CDT (geometric features +  
Concept Memory)

Decides *what kind* of drift it is (5 types).

## Layer 3 — Adaptation

Type-aware strategy

Decides *how* the model is updated.



# Contribution 1: Inverse Density-Weighted MMD

## Limitation of standard MMD (uniform weights):

- Dense majority clusters dominate the discrepancy term.
- Early drifts that appear at distribution boundaries are washed out by the bulk.

### Idea: weight by inverse local density

Down-weight points sitting in dense regions ( $w_i \downarrow$ ); amplify isolated points near the boundary ( $w_i \uparrow$ ). Boundary changes that signal drift get more influence on the test statistic.

**IDW-MMD:** weight  $w_i$  by inverse local density  $d_i$ .

## Weighted MMD form

$$w_i \propto \frac{1}{\sqrt{d_i + \epsilon}}, \quad d_i = \sum_j k(x_i, x_j)$$

$$\begin{aligned} \widehat{\text{MMD}}_{\text{IDW}}^2 &= \sum_{i,j} w_i w_j k(x_i, x_j) \\ &\quad + \sum_{p,q} v_p v_q k(y_p, y_q) \\ &\quad - \frac{2}{nm} \sum_{i,p} k(x_i, y_p) \end{aligned}$$

## Statistical calibration: Gamma-null p-value

The permutation test is replaced by a Gamma-approximation p-value (Gretton et al. 2009). This avoids re-computing the kernel

# Worked Example: 3-Point Window

A minimal observation window with 3 samples:

- Majority core:  $x_1, x_2$  sit close together (normal behaviour).
- Boundary point:  $x_3$  is isolated near the feature boundary (drift candidate).

## 1. Local density $d_i = \sum_j k(x_i, x_j)$

- Core:  $d_1 = k(x_1, x_1) + k(x_1, x_2) \approx 1.9$
- Core:  $d_2 = k(x_2, x_1) + k(x_2, x_2) \approx 1.9$
- Boundary:  $d_3 = k(x_3, x_3) + \text{small} \approx 1.0$

## 2. Inverse weights $w_i = 1/\sqrt{d_i}$

- Core:  $w_1 = w_2 = 1/\sqrt{1.9} \approx 0.72$  (down-weighted).
- Boundary:  $w_3 = 1/\sqrt{1.0} = 1.0$  (kept at full weight).

### Take-away

Standard MMD treats every point equally ( $w_i = 1$ ). IDW gives the boundary point ( $w_3 = 1.0$ ) more weight than the bulk ( $w_{1,2} = 0.72$ ), so a small change at the boundary moves the test statistic. This is what gives early sensitivity to drift while keeping false positives low on stationary data.

# Contribution 2: Unsupervised Shape Analysis of $\sigma(t)$

How can we type drift without ground-truth labels?

Read the *shape* of the MMD signal  $\sigma(t)$  around the detection point. SE-CDT extracts **9 statistical features**:

- **Geometric (peak structure):**

- $n_p$ : number of local maxima.
- WR (Width Ratio): peak width at half maximum.
- SNR: peak prominence over baseline noise.

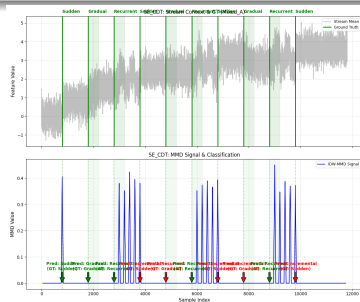
- **Temporal (behaviour over time):**

- LTS: linear trend strength of  $\sigma(t)$ .
- MS: monotonicity score (Mann-Kendall).

## Why these features?

Each drift type has a typical signature:

- **Sudden**: a single sharp peak ( $n_p = 1$ , WR small, SNR high).
- **Gradual**: no clean peak, baseline rises into a plateau.
- **Incremental**: sustained upward trend (LTS high, MS high).
- **Blip**: a quick dual-peak ( $n_p = 2$  with high amplitude ratio).



Hình: Sudden drift signature:  $n_p = 1$ , WR narrow.

# SE-CDT Decision Tree — Worked Example

## Rule-based decision tree:

| Condition                               | Label              |
|-----------------------------------------|--------------------|
| $n_p \leq 3$ , $WR < 0.15$ , $SNR > 2$  | <b>Sudden</b>      |
| $n_p = 2$ , $PPR < 0.2$ , $DPAR > 0.65$ | <b>Blip</b>        |
| MMD match in Concept Memory             | <b>Recurrent</b>   |
| $LTS > 0.5 \vee MS > 0.6$               | <b>Incremental</b> |
| Otherwise                               | <b>Gradual</b>     |

## Trace of one detection event:

Layer 1 has just fired a drift alert at time  $t^*$ .

- ① Read the MMD trace  $\sigma(t)$  over a window of  $2l_1$  samples around  $t^*$ .
- ② Count local maxima  $\rightarrow n_p = 1 (\leq 3)$ .
- ③ Width at half maximum  $\rightarrow WR = 0.12 (< 0.15)$ .
- ④ Peak vs baseline noise  $\rightarrow SNR = 3.0 (> 2)$ .

$\Rightarrow$  First rule matches  $\rightarrow$  label = **Sudden Drift**, which is then handed to the Adaptation layer.

# Hybrid Pipeline: IDW-MMD to Detect, Standard MMD to Classify

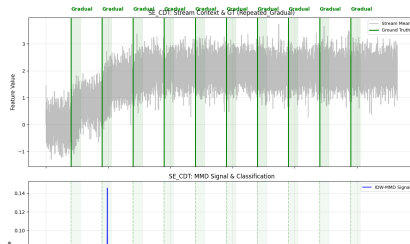
## Why use two MMD variants instead of one?

- **IDW-MMD** is sharper and almost never fires on stationary windows — great as a gatekeeper, but its weighting also *flattens* the slow rise of a Gradual drift, removing the very shape SE-CDT needs.
- **Standard MMD** runs slower but *preserves the natural geometry* of  $\sigma(t)$ , including plateaus and trends — ideal as input to the shape classifier.

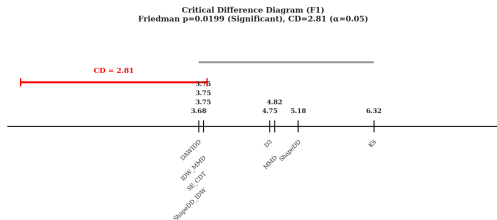
## Pipeline

- **Layer 1 — Detection:** IDW-MMD runs continuously as the gatekeeper.
- **Layer 2 — Classification:** Standard MMD is computed *only at the detection time  $t^*$*  to feed SE-CDT.

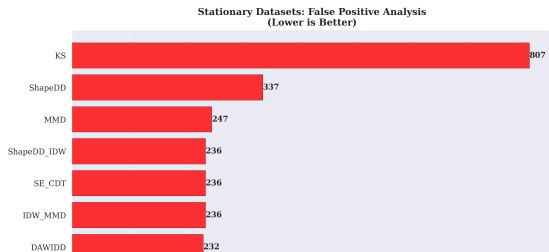
→ Cheap to run continuously, accurate where it matters.



# Result 1: Drift Detection (30 runs, 14 datasets)



Hình: Critical Difference Diagram (Nemenyi,  $\alpha = 0.05$ )



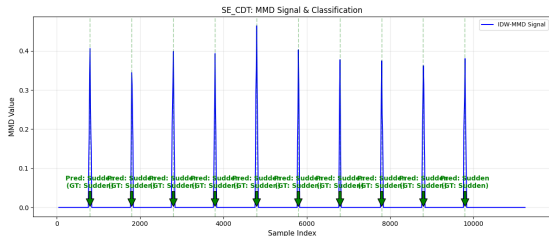
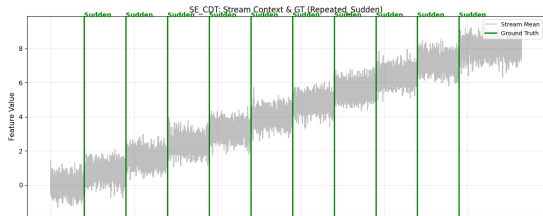
## Compared against 7 baselines:

| Method         | F1           | Prec. | FP   |
|----------------|--------------|-------|------|
| DAWIDD         | <b>0.532</b> | 0.436 | 10.3 |
| <b>IDW-MMD</b> | 0.531        | 0.432 | 9.6  |
| <b>SE-CDT</b>  | 0.531        | 0.432 | 9.6  |
| MMD (baseline) | 0.524        | 0.426 | 10.6 |
| D3             | 0.489        | 0.558 | 0.6  |
| KS-Test        | 0.335        | 0.224 | 27.2 |

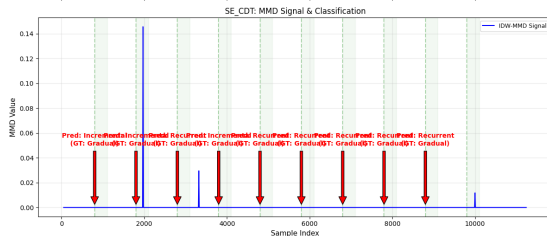
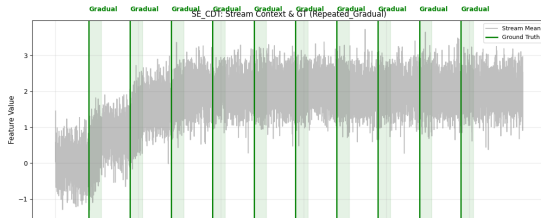
## Key Results

- Empirical Type-I error stays inside the 95% CI of  $\alpha = 0.05$  ( $H_0$ -calibration test).
- ShapeDD-IDW is faster than ShapeDD in the end-to-end benchmark (2.60 vs. 5.50 s per 10k-sample stream).
- On label-dominant drift datasets,  $F_1$  remains low (expected for an unsupervised  $P(Y)$ ).

# Drift Signals at the Detector Output

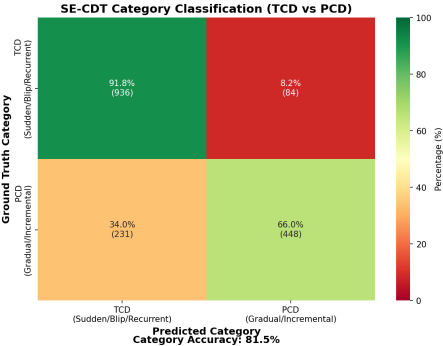


**Hình: Sudden drift:** sharp triangular peak — matches the Triangle Shape Theorem.



**Hình: Gradual drift:** IDW signal flattens (lower) while Standard MMD keeps the slope (upper) — motivates the hybrid pipeline.

# Result 2: Unsupervised Classification (SE-CDT)



Compared with CDT-MSW (supervised baseline):

|        | SE-CDT        | CDT-MSW  | Note                       |
|--------|---------------|----------|----------------------------|
| Labels | None          | Required | Unsupervised win           |
| CAT    | 81.5%         | 87.0%    | Close at -5.5pp            |
| SUB    | 65.6% / 54.0% | 86.7%    | micro / macro              |
| Recall | 92.6%         | 82.5%    | 1 - MDR from paper Table 5 |

Per-type accuracy (oracle mode, 1800 events):

| Drift Type  | Acc.  | Notes                             |
|-------------|-------|-----------------------------------|
| Sudden      | 82.4% | Sharp, well-separated peaks       |
| Recurrent   | 89.0% | Enabled by Concept Memory         |
| Incremental | 65.6% | Helped by temporal features       |
| Gradual     | 32.9% | Window signal returns to baseline |

Hình: Category Matrix (TCD vs PCD) — CAT = 81.5%

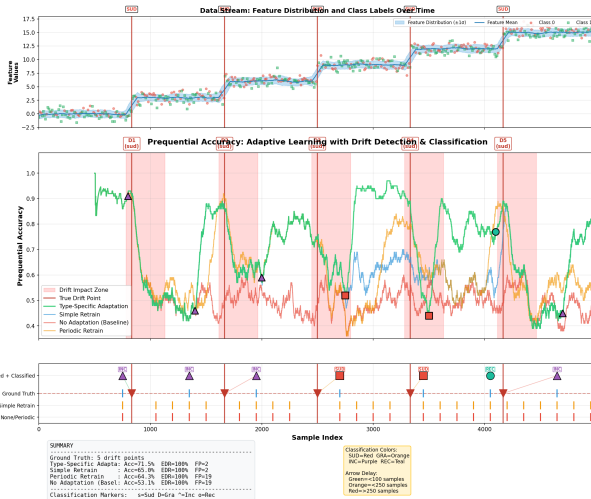
## Two enhancements added

**Concept Memory:** ring buffer of post-drift snapshots, matched via Standard MMD. Recovers Recurrent classification when the same concept reappears.

**Online Self-Calibration:** rolling quantile baseline that adapts



# Result 3: Model Adaptation (Prequential Accuracy, $N = 30$ )



| Strategy             | Step.        | Sudd.        |
|----------------------|--------------|--------------|
| <b>Type-Specific</b> | <b>71.4%</b> | <b>70.8%</b> |
| Periodic Retrain     | 64.0%        | 63.7%        |
| Simple Retrain       | 63.0%        | 62.2%        |
| No Adaptation        | 54.8%        | 53.7%        |

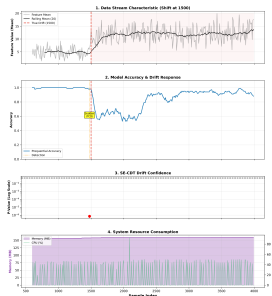
## Why Type-Specific helps

A single “always reset” policy (Simple Retrain) discards still-useful history when the drift is gradual; a fixed schedule (Periodic) ignores the drift signal altogether.

Routing the update strategy by drift type — *Sudden* → *full reset*, *Gradual* → *warm-start* — adds **+16.6 pp** on Stepping and **+17.1 pp** on Sudden over No Adaptation, with Friedman  $p < 10^{-15}$ .

**Hinh:** Stepping Drift: Type-Specific (blue) holds 71.4%, No Adaptation (red) drops to 54.8% as the distribution keeps moving away.

# End-to-End Deployment on Apache Kafka



## Detection

Drift alert fired at sample 1476, ahead of the planted change point at 1500.

## Recovery

Model accuracy recovers from 59% to **93%** after the type-aware adaptation step.

## Stability

Producer/detector/consumer run without back-pressure — closed-loop runs end-to-end.

# Conclusion & Future Directions

## Contribution Summary

- 1 **Detection:** ShapeDD-IDW with Gamma-null p-value  $\rightarrow$  a calibrated test ( $\alpha = 0.05$ , H0-verified) with a modest end-to-end speedup over ShapeDD.
- 2 **Classification:** SE-CDT label-free, CAT = **81.5%**, SUB = **65.6%** (micro), EDR = **92.6%**.
- 3 **Concept Memory + Self-Calibration:** restore Recurrent classification and stabilise SNR/WR/CV thresholds online.
- 4 **Adaptation:** Type-Specific gains **+17.1 pp** on Sudden, **+16.6 pp** on Stepping vs No Adaptation ( $N = 30$ , Friedman  $p < 10^{-15}$ ).
- 5 **System:** end-to-end Kafka closed-loop validated.

## Limitations

- Gradual/Incremental still hard to separate — sliding-window MMD returns to baseline.
- Concept Memory threshold  $\mu = 0.15$  chosen empirically from H0 noise floor.
- Adaptation: 3 scenarios,  $N = 30$  runs (Friedman  $p < 10^{-15}$ ).

## Future Outlook

- Multi-scale reference window for Gradual/Incremental.
- Bayesian decision boundaries (confidence scores).
- Concept Memory v2 with learned threshold.
- Real-world IoT validation.

**THANK YOU  
FOR LISTENING!**

**Q & A**

*Student: Le Phuc Duc — Supervisor: Assoc.Prof.Dr. Thoai Nam*

# Backup: IDW-MMD vs Optimally-Weighted MMD (Bharti et al.)

**Question:** “How is IDW-MMD different from the Optimally-Weighted MMD of Bharti et al. (2023)?”

## Bharti et al. (OW-MMD)

- Domain: **likelihood-free inference** (ABC).
- Weights chosen by optimising against a target posterior.
- Requires solving a constrained convex optimisation per call.
- Not designed for streaming drift detection.

## IDW-MMD (this work)

- Domain: **drift detection** in fast data streams.
- Weights from a closed-form heuristic:  
 $w_i \propto 1/\sqrt{d_i + \epsilon}$ .
- Cost:  $O(n^2)$  for the kernel matrix,  $O(n)$  extra for the weights — no optimisation step.
- Same *weighted MMD* family, different objective and very different runtime.

**Summary:** both are weighted MMD variants, but they target different problems (ABC vs streaming drift), use different weight functions, and have very different runtime profiles.

# Backup: “F1 close to MMD baseline — where is the contribution?”

**Question:** “SE-CDT’s F1 (0.531) is only marginally above MMD (0.524). Where is the gain?”

## F1 alone hides the trade-off:

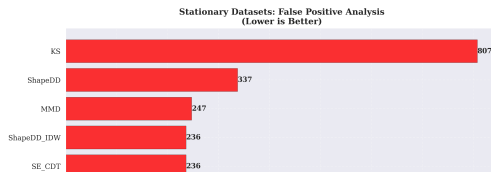
|           | SE-CDT     | MMD   | KS    |
|-----------|------------|-------|-------|
| F1        | 0.531      | 0.524 | 0.335 |
| <b>FP</b> | <b>9.6</b> | 10.6  | 27.2  |
| Recall    | 0.737      | 0.732 | 0.775 |
| Precision | 0.432      | 0.426 | 0.224 |

## **In a production setting:**

- Frequent FPs → **alert fatigue**, lost trust in the monitor.
- Investigating one false alarm typically costs more than missing one event.
- A miscalibrated permutation-style test inflates either FPs or misses.

## **What SE-CDT actually adds:**

- ➊ **Properly calibrated test** ( $\alpha = 0.05$ ) — empirical Type-I error inside the 95% CI on 3 stationary distributions.
- ➋ **Classification module on top of detection** — gives a drift type, not just a flag.
- ➌ **Modest runtime speedup over ShapeDD** thanks to the Gamma-null p-value (no kernel re-build per shuffle).
- ➍ Stays **fully unsupervised** at runtime.



# Backup: Why does Concept Memory matter for Recurrent drift?

**Question:** “Recurrent drift requires concept matching — how does Concept Memory provide it?”

## The detection challenge:

- A Recurrent stream is a sequence of *repeated* concept switches; each switch produces an MMD peak that, in isolation, looks like a Sudden drift.
- Without memory of past concepts, repeated visits are scored independently → Recurrent collapses into the Sudden branch.

## How Concept Memory resolves it:

- 1 During every *stable period*, take a snapshot of the current data window and store it in a ring buffer of size  $K = 8$ .
- 2 When a new drift fires, compute Standard MMD between the new post-drift window and every snapshot.
- 3 If the smallest  $\text{MMD} < \mu = 0.15 \rightarrow$  override the

### Why $\mu = 0.15$ ?

$\mu = 0.05$  falls inside the MMD noise floor (false matches between independent stationary windows). An empirical sweep on H0 calibration data shows  $\mu = 0.15$  separates “revisited concept” from “new concept” reliably.

### Remaining limitations

Two failure modes remain on the harder Recurrent variants:

- Recurrent events that are *too far apart* push the original snapshot out of the ring buffer.
- Repeated Gradual streams produce flattened MMD signals that no snapshot matches well.

# Backup: “CDT-MSW has higher CAT and SUB — is SE-CDT worse?”

**Question:** “CDT-MSW reports CAT 87.0% and SUB 86.7%, both higher than SE-CDT. Why frame SE-CDT as competitive?”

## Side-by-side numbers

|           | SE-CDT       | CDT-MSW  |                   |
|-----------|--------------|----------|-------------------|
| Labels    | <b>None</b>  | Required |                   |
| CAT       | 81.5%        | 87.0%    | Effective = CAT × |
| SUB       | 65.6%        | 86.7%    |                   |
| Recall    | <b>92.6%</b> | 82.5%    |                   |
| Effective | <b>75.5%</b> | 71.8%    |                   |

Recall (probability of seeing a drift *and* typing it correctly).

CDT-MSW Recall =  $1 - \overline{\text{MDR}}$  from paper Table 5.

## Reading the numbers in context:

- CDT-MSW reports CAT and SUB *conditionally* on drifts it detects; with paper recall  $\approx 82.5\%$  it still misses  $\sim 17\%$  of events.
- SE-CDT detects 92.6% of drifts and types every one of them without any labels.
- End-to-end, SE-CDT lands  $\sim 4.0\text{pp}$  above CDT-MSW on effective accuracy (75.8% vs 71.8%); the contribution is reaching this on **unsupervised**  $P(X)$ .

## Trade-off

CDT-MSW is more accurate per detected event because it has access to ground truth labels. SE-CDT



# Backup: “Why not deep learning?”

**Question:** “Couldn’t a deep learning model solve this more accurately?”

## Deep models for drift detection — limitations

- Need a large labelled dataset to train — the very thing we don’t have on a streaming IIoT pipeline.
- Higher inference latency per window than a kernel computation.
- No closed-form theoretical bound on the test statistic.
- Decision is hard to explain to a domain expert.

## Why MMD / ShapeDD fits this problem

- Fully unsupervised — no training run before deployment.
- Theoretical guarantees: Triangle Shape Theorem, Type-I error from  $H_0$  calibration.
- Decision rule is human-readable (a small tree of geometric features).
- One mechanism works across  $P(X)$  shifts of arbitrary type.

**Future direction:** keep the kernel framework but plug in a learned feature extractor for high-dimensional inputs (Chapter 5).

# Backup: weakness on mild drift

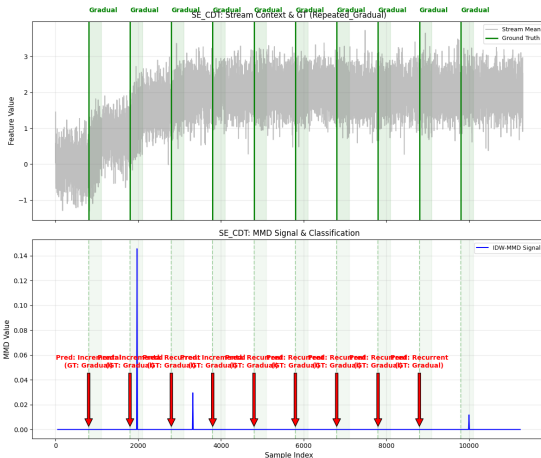
**Question:** “On the mild-drift scenario SE-CDT scored  $F1 \approx 0.006$  — why so low?”

## Cause:

- IDW down-weights points sitting in dense (stable) regions.
- Mild drift, by definition, lives mostly inside those dense regions.
- Net effect: the IDW signal is too smooth to cross threshold (Standard MMD reaches  $\sim 0.585$  on the same data).

## This is the intended trade-off:

- IDW-MMD biases toward high precision (FP  $\sim 2/\text{run}$ , lowest among baselines).
- Cost: lower recall on very small distributional shifts.



**Hình:** IDW signal (lower) vs Standard MMD (upper) on the same

# Backup: Permutation test vs Gamma-null p-value

**Question:** “Removing the permutation test — is the Gamma-null p-value still trustworthy?”

## Permutation test (original ShapeDD)

- Reshuffles indices many times per window.
- Empirical null, no distributional assumption.
- Each shuffle re-builds the kernel matrix  $\rightarrow$  heavy compute.

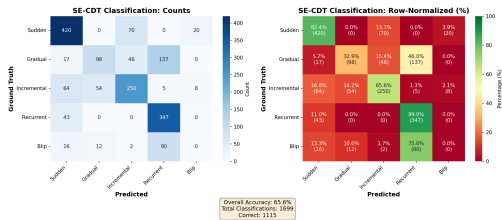
## Gamma-null p-value (IDW-MMD)

- Moment-matched Gamma fit to the null MMD distribution (Gretton et al. 2009).
- Estimates the first two moments from a small number of index shuffles, reusing the kernel matrix.
- Complexity  $O(n^2)$  for the kernel plus a few cheap shuffles.

**Verification:** The detection  $F_1$  remains comparable to the permutation baseline, while H0-calibration on three stationary distributions reports an empirical Type-I error of 0.04–0.10, inside the 95% CI of  $\alpha = 0.05$ . End-to-end runtime is slightly faster than the ShapeDD baseline.

# Backup: “Subcategory accuracy 65.6% — is it close to random?”

**Question:** “With 5 classes, 65% sounds close to random guessing.”



Hình: 5-class confusion matrix (SE-CDT)

**It is well above random:**

- Random baseline across 5 classes is 20%.
- Micro SUB = 65.6% is  $\sim 3.3\times$  random and well above the 95% CI of a random predictor ( $\sim 23\%$ ).
- Performance is **very uneven across classes**:

| Drift Type | Accuracy |
|------------|----------|
|------------|----------|

|           |       |
|-----------|-------|
| Recurrent | 89.0% |
|-----------|-------|

|        |       |
|--------|-------|
| Sudden | 82.4% |
|--------|-------|

|             |       |
|-------------|-------|
| Incremental | 65.6% |
|-------------|-------|

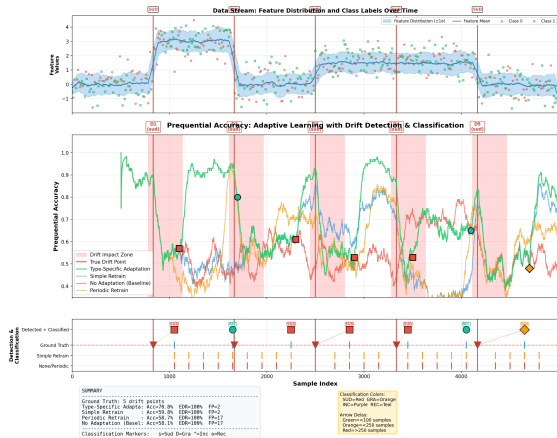
|         |       |
|---------|-------|
| Gradual | 32.9% |
|---------|-------|

|      |      |
|------|------|
| Blip | 0.0% |
|------|------|

**Why the gap?** Recurrent benefits from Concept Memory; Sudden has the cleanest peaks. Incremental and Gradual look similar at the

# Backup: "Is the adaptation gap statistically significant?"

**Question:** "Is Type-Specific really better, or is it run-to-run noise?"



**Friedman test (paired by seed):**

- Stepping:  $\chi^2 = 78.12$ ,  $p = 7.8 \times 10^{-17}$
- Sudden:  $\chi^2 = 86.76$ ,  $p = 1.1 \times 10^{-18}$
- Mixed:  $\chi^2 = 76.32$ ,  $p = 1.9 \times 10^{-16}$

$p \ll 0.05$  on all three scenarios; the ranking among 4 strategies is not random.

## Remaining limits

- Only 3 scenarios (Stepping, Sudden, Mixed).
- No Gradual/Incremental adaptation scenario yet.
- No real-world dataset with verified drift labels.

Listed in **Chapter 5: Future Directions.**

**Hinh:** Prequential accuracy on Sudden drift,  $N = 30$  runs.